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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2026

Code: 104602		***Time: 03 Hours***
Semester: VI	***Subject: Digital Communication***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) In a PCM system with uniform quantization, increasing the number of bits from 8 to 9 will reduce the quantization noise power by a factor of

(i) 9
(ii) 4
(iii) 8
(iv) 2

(b) Which one of the following is an analog system?

(i) PCM
(ii) Delta modulation
(iii) Differential PCM
(iv) PAM

(c) The main advantage of DPCM over conventional PCM is

(i) Higher quantization error
(ii) Reduced number of bits required for a given quality
(iii) Elimination of sampling
(iv) Elimination of encoding

(d) Non-coherent FSK detection has the advantage that it

(i) Does not require phase synchronization with the carrier
(ii) Requires perfect carrier phase recovery
(iii) Always has zero error probability
(iv) Requires no receiver

(e) In flat-top sampling, the sampled pulse

(i) Changes continuously during the sampling interval
(ii) Has zero amplitude
(iii) Maintains a constant amplitude during the sampling interval
(iv) Has a triangular shape

(f) The primary purpose of a matched filter is to

(i) Maximize signal power at all frequencies
(ii) Eliminate the carrier
(iii) Maximize output SNR at a specified sampling instant
(iv) Increase bandwidth

(g) If the available bandwidth is divided among several FDM users, increasing the number of users generally results in

(i) More bandwidth per user
(ii) Less bandwidth per user
(iii) Zero total bandwidth
(iv) Elimination of interference

(h) The raised-cosine spectrum is mainly used for

(i) Increasing quantization noise
(ii) Controlling bandwidth while satisfying the zero-ISI criterion
(iii) Removing the transmitter
(iv) Generating thermal noise

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(i) An analog signal contains frequency components from 2 kHz to 5 kHz. If it is sampled at 6 kHz, which problem is most likely to occur?

(i) Quantization noise
(ii) Thermal noise
(iii) Aliasing
(iv) Inter-symbol interference

(j) Adaptive delta modulation reduces slope-overload distortion by

(i) Decreasing sampling frequency
(ii) Increasing the step size when the signal changes rapidly
(iii) Eliminating quantization
(iv) Removing the feedback path

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Derive the expression for probability of error for QPSK signal.

(b) Represent the data stream '101110' using NRZ, RZ, Manchester, AMI and Bipolar line coding techniques.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) If a voice frequency signal is sampled at the rate of 32,000 *samples/sec* and characterized by peak value of 2 volts, determine the value of step size to avoid slope overload. What is quantization noise power and corresponding SNR? Assume bandwidth of signal as 4 kHz.

(b) Define probability of error and explain factors affecting BER in digital communication systems.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain Binary Phase Shift Keying generation and coherent demodulation.

(b) State and explain the sampling theorem. Discuss the conditions that must be satisfied for faithful reconstruction of a band-limited signal.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Draw and explain the QPSK constellation diagram.

(b) Explain the generation and detection process in CDMA systems.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain Adaptive Delta Modulation and its advantages over conventional Delta Modulation.

(b) Consider a random process $x(t) = 3v(t) - 8$, where $v(t)$ is a zero mean stationary random process with $R_v(\tau) = 4e^{-5|\tau|}$. Calculate the power in $x(t)$.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Find the Nyquist rate and the Nyquist interval for the signal $x(t) = 1/(2\pi) \cos(4000\pi t) \cos(1000\pi t)$

(b) Derive the probability of error for coherent Binary FSK and Binary PSK.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) What is aperture effect and explain how to reduce it.

(b) Explain non-uniform quantization using companding techniques. Discuss μ-law and A-law companding.

Q.9 Write short notes on any two of the following: **[7 x 2 = 14]**

(a) ASK, FSK and PSK

(b) Inter-Symbol Interference

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(c) Frequency division multiplexing

(d) Matched Filter

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